

SOFTWARE QUALITY ASSURANCE REPORT

Prepared by	Senior QA Engineer / Test Architect
Date	2026-05-11
Repository	AbuRashad/SchoolSmartEYE
Test Environment	Python 3.12.3 · pytest 8.4.2 · FastAPI TestClient · OpenCV

EXECUTIVE SUMMARY

Metric	Result
Test Suite Pass Rate	■ 100% (93 / 93 tests)
API Endpoint Coverage	29 / 31 routes functional (93.5%)
Critical Bugs Found	2 confirmed + 2 design concerns
Security Controls	■ Implemented & Verified at service layer
Real-Time WebSocket	■ Fully Functional
Deployment Readiness	■■ Phased Rollout Recommended

SECTION 1 — TEST SUITE ANALYSIS

1.1 Coverage by Domain

Test Module	Tests	Pass	Fail	Coverage Notes
test_health.py	1	1	0	Health endpoint smoke test
test_dashboard.py	4	4	0	Summary, alerts, heatmap, WebSocket
test_governance.py	2	2	0	RBAC scope isolation
test_school_safety_index.py	7	7	0	SSI math, weight normalization, clipping
test_camera_ingestion.py	8	8	0	Pipeline, file loading, edge cases
test_ministry_report_service.py	4	4	0	SSI accuracy, anonymization, standards
test_parent_portal_service.py	3	3	0	Access control, notifications, privacy
test_pedagogical_agent.py	37	37	0	Largest suite — state, scoring, history, WS
test_report_generator.py	3	3	0	Avg score, top risk zones, format
test_risk_heatmap_generator.py	3	3	0	Data fusion, temporal shift, overlay
test_collective_behavioral_coherence.py	3	3	0	Alignment/divergence, entropy, rupture
test_crowd_density_forecasting.py	3	3	0	Normal/peak patterns, proactive alerts
test_spatial_behavioral_memory.py	2	2	0	Initialization, anomaly detection
test_drill_simulation_service.py	3	3	0	Scenario build, scoring, data isolation
test_attendance_safety_integration.py	3	3	0	Zone match, restricted zone, missing
TOTAL	93	93	0	■ 100% pass rate — run time: 2.71 s

1.2 Test Quality Assessment

Dimension	Assessment
Positive path coverage	■ PASS — All happy paths covered
Negative / edge case coverage	■ PASS — Invalid JSON, missing files, out-of-range scores
Boundary value testing	■ PASS — Score clipping [0-1], weight normalization, cutoffs
Test isolation	■ PASS — All in-memory; no shared DB, no side effects
Async coverage	■ PASS — asyncio tests for all WebSocket and async HTTP routes
Contract testing	■ PASS — Response schema keys validated (not only status codes)
Integration tests	■■ MISSING — No test covers full camera→SSI→alert→WS push flow
Load / stress tests	■■ MISSING — No concurrency or throughput tests
Frontend E2E tests	■■ MISSING — No Playwright/Cypress tests for React frontend
Negative auth tests	■■ MISSING — No test attempts unauthorized HTTP access

SECTION 2 — API ENDPOINT TEST RESULTS

2.1 Functional Verification (34 documented routes)

Route	Method	HTTP	Result
/api/v1/health	GET	200	■ PASS
/api/v1/dashboard/summary	GET	200	■ PASS
/api/v1/dashboard/alerts	GET	200	■ PASS
/api/v1/dashboard/heatmap	GET	200	■ PASS
/api/v1/dashboard/live	GET	200	■ PASS
/api/v1/dashboard/ws	WS	101	■ PASS
/api/v1/ssi/live	GET	200	■ PASS
/api/v1/ssi/history	GET	200	■ PASS
/api/v1/units	GET	200	■ PASS
/api/v1/units/capture/health	GET	200	■ PASS
/api/v1/units/alerts/summary	GET	200	■ PASS
/api/v1/analytics/overview	GET	200	■ PASS
/api/v1/reports/list	GET	200	■ PASS
/api/v1/reports/stats	GET	200	■ PASS
/api/v1/portal/student	GET	200	■ PASS
/api/v1/cameras	GET	200	■ PASS
/api/v1/cameras/{id}	GET	405	■ BUG
/api/v1/cameras/{id}/snapshot.jpg	GET	404	■■ Expected (no camera configured)
/api/v1/cameras/{id}/start	POST	—	■ PASS
/api/v1/cameras/{id}/stop	POST	—	■ PASS
/api/v1/students	GET	200	■ PASS
/api/v1/students/{id}	GET	200	■ PASS
/api/v1/students/{id}/profile	GET	200	■ PASS
/api/v1/students/{id}/behavioral-history	GET	200	■ PASS
/api/v1/bracelets	GET	200	■ PASS
/api/v1/bracelets/{id}	GET	405	■ BUG
/api/v1/bracelets/{id}/ping	POST	—	Defined
/api/v1/settings	GET	200	■ PASS
/api/v1/agent/insights	GET	200	■ PASS
/api/v1/agent/baselines	GET	200	■ PASS
/api/v1/agent/feedback	POST	201	■ PASS

/api/v1/agent/feedback/metrics	GET	200	■ PASS
/api/v1/zones	GET	200	■ PASS
/openapi.json	GET	200	■ PASS

Pass rate: 29 / 31 testable routes = 93.5%

SECTION 3 — REAL-TIME & STREAMING TESTS

3.1 WebSocket (/api/v1/dashboard/ws)

Test	Result	Detail
Initial connection	■ PASS	Snapshot delivered immediately on connect
refresh command	■ PASS	New computed snapshot returned
subscribe command	■ PASS	Equivalent snapshot returned
ping command	■ PASS	Full dashboard snapshot returned as pong
SSI value consistency	■ PASS	SSI=76 consistent across all WS responses
Alert count consistency	■ PASS	2 alerts (coherence + anomaly) in all responses
Heatmap cell count	■ PASS	8 cells (4 live + 4 prediction_15m)
fetchFromBackend flag	■ PASS	true on all responses

3.2 Camera Streaming

Feature	Result	Reason
Snapshot endpoint	404 (expected)	No camera registered in test environment
MJPEG stream	N/A	Requires RTSP hardware source
Camera registration	■ PASS	test_camera_ingestion.py covers full pipeline
Offline-until-first-frame	■ PASS	Status stays offline until frame arrives

SECTION 4 — CORE ALGORITHM TESTS

4.1 School Safety Index (SSI)

- Computation verified: SSI = 76.04 with live inputs (anomaly=0.18, coherence=0.76, attendance_discrepancy=0.08, density_risk=0.4384)
- Weight auto-normalization: ■ Tested — mismatched weights are normalized
- Score clipping: ■ Tested — out-of-range indicators are clipped to [0, 1]
- Benchmark comparison: ■ SSI correctly flagged as above_benchmark
- ■■ Note: Direct SSI from service = 63.04 (below benchmark) vs live endpoint = 76.04 — different input pipeline; divergence should be documented.

4.2 Crowd Density Forecaster

- CrowdDensityForecaster: Requires fit(historical_data) before predict() — raises clear RuntimeError ■
- Time-aware predictions: Different density values across hours 07:00–17:00 ■
- CrowdDensityLSTMForecaster: Requires fit() and training data before inference ■■ (not auto-initialized)

4.3 Collective Behavioral Coherence (CBC)

- Uniform/static motion: coherence=0.0, entropy=1.0, rupture=True on static input ■■ (zero flow = max entropy in Farneback model; expected by design, but needs documentation)
- Random motion: coherence=0.755, entropy=0.745, rupture=False ■

4.4 Spatial Behavioral Memory (SBM)

- 4×4 grid: 16 cells processed correctly ■
- Dynamic threshold: 0.65 (before learning baseline stabilizes) ■
- Anomaly coefficient: Computed from grid-cell Z-score deviation ■

4.5 Risk Heatmap Generator

- 16-cell fused grid: Anomaly weight=0.6, density weight=0.4 ■
- Both ingest_sbm_anomaly() and ingest_predictive_density() consume numpy grids ■

SECTION 5 — GOVERNANCE & SECURITY TESTS

5.1 Role-Based Access Control (RBAC)

Role	Scope Level	Allowed Actions	Test Result
Teacher	1	view_classroom_feed only	■ PASS
Principal	2	classroom + analytics + retention	■ PASS
Ministry	3	system-wide + all above	■ PASS
Teacher → other class	—	Blocked	■ PASS
Teacher → system analytics	—	Blocked	■ PASS
Ministry → any school	—	Allowed	■ PASS

5.2 Privacy Controls

Control	Test	Result
Minor face anonymization	Gaussian blur applied to all detected faces (conservative: no age estimator = blur all)	■ PASS
Video TTL enforcement	File older than 72h deleted; recent file preserved	■ PASS
camera_anonymize_faces=true	Default setting confirmed in /api/v1/settings	■ PASS
Ministry report anonymization	Aggregated-only output (no individual student data)	■ PASS
Parent portal privacy	Detailed scene analysis hidden from parent view	■ PASS

5.3 Security Gaps Identified

Gap	Severity	Detail
No HTTP auth on API routes	■ HIGH	All 34 API routes publicly accessible — no token/JWT enforcement
RBAC only at service layer	■ Medium	GovernanceLayer exists but not enforced as HTTP middleware
WebSocket: no auth	■ HIGH	WS connection accepts any client — no token validation
Agent feedback: no ownership check	■ Medium	Any caller can post feedback for any student ID

SECTION 6 — BUGS FOUND (DEFECT LOG)

■ BUG-001	Severity: Medium Type: Missing route handler
Actual	GET /api/v1/cameras/{camera_id} returns HTTP 405
Expected	HTTP 200 with camera detail object
Root Cause	Router defines POST /start, POST /stop but no GET for individual camera
Impact	Dashboard/frontend cannot fetch single-camera status

■ BUG-002	Severity: Medium Type: Missing route handler
Actual	GET /api/v1/bracelets/{bracelet_id} returns HTTP 405
Expected	HTTP 200 with bracelet detail + student + zone
Root Cause	Router defines bracelet list and POST /ping but no GET per bracelet
Impact	Cannot fetch single bracelet status for battery/location drill-down

■ BUG-003	Severity: Low Type: Undocumented behavioral difference
Actual	compute_school_safety_index() with identical inputs = 63.04 vs /api/v1/ssi/live = 76.04
Expected	Same SSI value through both paths for the same inputs
Root Cause	Live endpoint uses different simulated inputs from the service layer
Impact	Developers may incorrectly assume they are testing the same data path

■ BUG-004	Severity: Low Type: Data quality / cold-start concern
Actual	Pedagogical Agent feedback accuracy starts at 50% on fresh data
Expected	Accuracy >=80% before system is considered tuned
Root Cause	Insufficient training data; improvement note is correctly generated
Impact	Agent is usable but prediction quality is low at cold start

SECTION 7 — TEST GAPS & RECOMMENDATIONS

7.1 Missing Test Coverage

Gap	Priority	Recommendation
No HTTP auth enforcement test	■ HIGH	Add test verifying 401 on unauthenticated request
No WebSocket auth test	■ HIGH	Add test verifying WS rejection without valid token
No integration test (camera→SSI→WS)	■ MED	Add end-to-end pipeline test with fake video frame
No concurrency test	■ MED	Simulate 10+ WS clients simultaneously
No frontend E2E test	■ MED	Add Playwright tests for dashboard KPIs, heatmap, portal
No GET /cameras/{id} test	■ MED	Add test after BUG-001 is fixed
No GET /bracelets/{id} test	■ MED	Add test after BUG-002 is fixed
LSTM: no graceful-degradation test	■ LOW	Verify endpoint degrades gracefully without trained weights
No negative governance HTTP test	■ HIGH	Test that Teacher role cannot call ministry-level routes

7.2 Code Quality Observations

Observation	Impact
All state is in-memory (dicts/lists)	No persistence — restart loses all data; no database layer yet
seed_data.populate() is idempotent	■ Good practice — safe to call multiple times
Service modules use clean dataclasses	■ Good — type-safe, easily testable
CBC process_frame on static frames → rupture=True	Needs documentation to avoid operator confusion in production
LSTM requires manual fit() — no auto-init path	Should be initialized with historical data at startup

SECTION 8 — DEPLOYMENT READINESS SCORECARD

Module	Status	Notes
Health endpoint	■ Production-ready	
Dashboard (REST + WS)	■ Production-ready	
SSI service	■ Production-ready	Divergence with live endpoint needs documentation
Analytics overview	■ Production-ready	
Reports (list + stats)	■ Production-ready	
Parent portal	■ Production-ready	
Students API	■ Production-ready	
Bracelets (list)	■ Production-ready	
Bracelets (by ID)	■■ Needs fix (BUG-002)	GET handler missing
Settings	■ Production-ready	
Zones	■ Production-ready	
Governance / RBAC	■■ Needs HTTP middleware	Service-layer only; not enforced on router
Cameras (list)	■ Production-ready	
Cameras (by ID)	■■ Needs fix (BUG-001)	GET handler missing
Camera streaming	■■ Needs hardware config	Requires RTSP source — expected
Pedagogical Agent	■■ Needs tuning	Cold-start accuracy 50%; needs feedback volume
CBC anomaly detection	■■ Needs documentation	Counterintuitive behavior on static frames
LSTM density forecaster	■■ Not production-ready	Requires training data + model weights
Frontend (React)	■■ Not tested (no E2E)	No automated frontend tests exist

SECTION 9 — FINAL VERDICT & ROLLOUT PLAN

ED FOR PHASED ROLLOUT

Mandatory pre-conditions apply — see

Phase 1 — Immediate deploy (backend core)

- Deploy all 29 verified API routes: health, dashboard, SSI, analytics, reports, portal, students, bracelets (list), settings, zones.
- All routes are stable and production-quality.
- PRE-CONDITION: Add JWT authentication middleware before exposing to public network.

Phase 2 — 1–2 weeks

- Fix BUG-001 (GET /cameras/{id}) and BUG-002 (GET /bracelets/{id}).
- Register real RTSP cameras and validate snapshot/stream endpoints.
- Add HTTP auth enforcement tests (401 on unauthenticated access).

Phase 3 — 1 month

- Accumulate teacher feedback to push Pedagogical Agent accuracy above 80%.
- Feed historical density data to activate real CrowdDensityForecaster (not simulated).
- Add Playwright frontend E2E test suite covering dashboard, heatmap, and portal.

Phase 4 — Full rollout

- After one semester of live SSI tracking vs. 75-point benchmark.
- After passing load tests (50+ concurrent WebSocket clients).
- After security penetration test confirms auth middleware is fully effective.

Report Status	FINAL
Overall Test Confidence	HIGH on tested paths / MEDIUM on untested auth/security paths
Recommendation	■■ Do NOT expose to internet without auth middleware